Day 29 Deliverable: FinDoc-AuditEngine v3.0 System Architecture Specification

1. High-Level Component Topology

flowchart TB
subgraph Client_Layer ["1. Presentation & Client Layer"]
WEB["Institutional Web Workbench\n(index.html + dashboard.css + dashboard.js)"]
CLI["CLI Runners\n(run_project.py / run_mentor_demo.py)"]
SWAGGER["OpenAPI / Swagger UI\n(/docs)"]
end

subgraph API_Gateway ["2. FastAPI Application & Exception Layer (app/main.py)"]
ROUTER_API["REST Router (/api/v1/audit)\n- POST /process\n- POST /parse-raw\n- POST /demo-batch\n- GET /ledger & /summary\n- GET /export/csv"]
ERR_HANDLER["RFC 7807 Global Exception Handler\n(FinDocAuditException -> JSONResponse)"]
end

subgraph Core_Domain ["3. Core AI/ML & Reconciliation Services"]
PII["Stage 1: FinancialPrivacyScrubber\n(PAN, Aadhaar, Email, Phone, Account)"]
MATCH["Stage 2: ReconciliationMatcher\n(Line-Item Invoice vs PO vs Dock GRN)"]
ML["Stage 3: AnomalyScorer\n(7D StandardScaler + IsolationForest + Perturbation XAI)"]
GOV["Stage 4: ApprovalRouter\n(5 Governance Tiers + Dual USD/INR @ 86.50)"]
end

subgraph Storage_Layer ["4. State & Compliance Export"]
LEDGER["Immutable Audit Ledger & PO Registry"]
CSV_EXP["CSV Compliance Report Generator"]
end

WEB --> ROUTER_API
CLI --> ROUTER_API
SWAGGER --> ROUTER_API
ROUTER_API --> ERR_HANDLER
ROUTER_API --> PII
PII --> MATCH
MATCH --> ML
ML --> GOV
GOV --> LEDGER
LEDGER --> CSV_EXP

2. 7-Dimensional ML Feature Vector Specification

Index	Feature Name	Mathematical Formulation	Normal Operational Range	Anomaly Trigger Condition
0	Log Capital Scale	$\ln(1 + \text{Total}_{\text{USD}})$	$4.8 - 10.3$ ($120 - $30k)	Extreme capital outliers (> $40,000)
1	Unit Rate Markup %	$\frac{\|\Delta P_{\text{USD}}\|}{\text{Base Spend}} \times 100$	$0.0\% - 0.5\%$	Vendor unit price exceeds PO rate (> 1%)
2	Dock GRN Shortage %	$\frac{\Delta Q_{\text{units}}}{Q_{\text{billed}}} \times 100$	0.0%	Billed units rejected or missing at dock
3	Tax Deviation	$\max(0, \text{Tax}\% - 20.0\%)$	0.0%	Tax rate exceeds corporate indirect tax ceiling
4	Payment Urgency	$\min(10, \frac{30}{\max(1, \text{DueDays})})$	$1.0 - 2.0$ (Net-30 to Net-15)	Rush wire demand (<= 5 days, urgency >= 6.0)
5	Vendor Risk Profile	$0.55 \cdot \mathbb{I}_{\text{new}} + 0.45 \cdot \mathbb{I}_{>\$5k} + 0.75 \cdot \mathbb{I}_{\neg\text{PO}}$	$0.0 - 0.35$	New vendor high-value spend or missing PO
6	Jurisdiction AML Index	Lookup in JURISDICTION_RISK_MAP	$0.05 - 0.08$ (IN, US, DE, GB)	Offshore tax havens (BZ, KY, PA, VG ≥ 0.88)

3. Corporate Governance Routing Matrix (1 USD = 86.50 INR)

Approval Tier	USD Threshold	INR Threshold (@ 86.50)	3-Way Match Required	Max Risk Level	HITL Required
TIER_1_STP_AUTO_APPROVED	< $1,000.00	< INR 86,500.00	True (matched)	LOW (< 35.0)	False (Autonomous)
TIER_2_MANAGER_REVIEW	$1,000 – $10,000	INR 86,500 – 865,000	True (matched)	LOW / MEDIUM	True (Manager)
TIER_3_DIRECTOR_SIGNOFF	> $10,000.00	> INR 865,000.00	True (matched)	LOW / MEDIUM / HIGH	True

(Director) |
| REJECTED_DISCREPANCY | Any Amount | Any Amount | False (unmatched) | Any | True (Vendor Hold) |
| FROZEN_FRAUD_RISK | Any Amount | Any Amount | Any | CRITICAL (>= 85.0) | True (Security Freeze) |